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COURSE: Machine Learning

TASK1: IRIS CLASSIFICATION

NOTEBOOK LINK: https://colab.research.google.com/drive/1cGmKtJhVVa9yvnQGhYor-pJLdPUtOsrr?usp=sharing

PLOTS:

- Pair Plot
- Heatmap
- Scatter Plot
- Confusion Matrix
- Accuracy Comparison Bar Chart

Saved Model Files:

best_model.pkl → trained ML model

scaler.pkl → feature scaling object

README with inference example:

Iris Species Classification

Objective

Build a machine learning model to classify iris flower species using:

- Sepal length

- Sepal width

- Petal length

- Petal width

Algorithms Used

- K-Nearest Neighbors (KNN)

- Logistic Regression

- Decision Tree

Dataset

Scikit-learn Iris Dataset

Steps Performed

1. Data Loading
2. Exploratory Data Analysis (EDA)
3. Data Visualization
4. Feature Scaling
5. Model Training
6. Model Evaluation
7. Model Saving

Evaluation Metrics

- Accuracy
- Confusion Matrix
- Precision
- Recall
- F1-score

Best Model

K-Nearest Neighbors (KNN)

Saved Files

- best_model.pkl

- scaler.pkl

Example Inference Code

```python
import joblib

model = joblib.load('best_model.pkl')
scaler = joblib.load('scaler.pkl')

sample = [[5.1, 3.5, 1.4, 0.2]]

sample = scaler.transform(sample)

prediction = model.predict(sample)

print(prediction)
```

Executive Summary :

The Iris Flower Classification project is a machine learning-based application developed to classify iris flowers into three different species: Iris-setosa, Iris-versicolor, and Iris-virginica. The project uses the famous Iris dataset, which contains measurements such as sepal length, sepal width, petal length, and petal width. The main objective of this project is to build an accurate and efficient predictive model using machine learning algorithms and to understand the complete workflow of a data science project.

The project begins with importing essential Python libraries such as NumPy, Pandas, Matplotlib, Seaborn, and Scikit-learn. The dataset is loaded and explored using data analysis techniques to understand the structure, distribution, and relationships between features.

Various visualization techniques including pair plots, scatter plots, histograms, box plots, and heatmaps are used to analyze the data and identify patterns among the flower species.

Data preprocessing techniques are applied to prepare the dataset for machine learning. Feature scaling is performed using StandardScaler to improve model performance. The dataset is then divided into training and testing sets for proper evaluation of the model. Multiple machine learning algorithms such as Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, and Random Forest are implemented and compared based on their accuracy scores.

Among all models, the K-Nearest Neighbors algorithm achieved the best performance with high classification accuracy. The trained model is saved using Joblib as a `.pkl` file, allowing the model to be reused later without retraining. The scaler object is also saved for future predictions and deployment purposes.

The project demonstrates important concepts of machine learning including data preprocessing, exploratory data analysis, model training, evaluation, prediction, and model persistence. It provides practical experience in developing predictive systems and helps in understanding how machine learning can be applied to real-world classification problems.

Overall, the Iris Flower Classification project successfully achieves accurate flower species prediction while demonstrating a complete end-to-end machine learning workflow using Python and Scikit-learn.

GITHUB repo Link: **https://github.com/Subhasree2626/Alfido-tech_Iris-Classification.git**